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Beyond geopolitics: Social license and supply chain risks of critical minerals

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ABSTRACT

This paper interrogates the concept of criticality in critical minerals, particularly as it pertains to the global supply chains essential for renewable energy and clean technologies. It begins by questioning whose interests define mineral criticality, noting that dominant perspectives (i.e., industry, state, and geopolitics) often marginalize sub-national concerns. A fourth dimension, the role of communities and the presence or absence of social license to operate, is widely studied in mining literature but rarely integrated into criticality assessments. Drawing on qualitative data and a case study of Ghana's emerging lithium sector, the paper reframes criticality to include sub-national social dynamics and offers a critical reflection on how social license functions as a risk factor in critical minerals governance. The findings reveal that delays in legal processes, exclusion from decision-making, and inadequate compensation disrupt livelihoods and heighten tensions in lithium-rich communities. These dynamics elevate the risk of social conflict, which in turn increases the cost of conflict for both state and companies. The paper argues that effective risk governance must embed principles of Free, Prior and Informed Consent, equitable beneficiation, and sustained community engagement. Social license to operate is not a static condition but a dynamic process that can be withdrawn, with significant implications for project timelines, reputational risk, investor confidence, and supply chains risks. As global competition intensifies over access to strategic minerals, the governance of mining sites in the Global South becomes important for supply chain assurance. The findings underscore the importance of integrating social dimensions into global criticality frameworks.

1. Introduction

Critical minerals have become synonymous with green minerals in contemporary discourses on energy transitions. They are widely used to characterise the low-volume minerals necessary to power the renewable energy systems in everyday technology such as wind turbines, solar systems, and electric vehicles, and in strategic defence technology such as green submarines. Critical minerals' use advances the Paris Agreement and Sustainable Development Goals (SDGs) 7 and 13 that aim to attain net zero carbon emissions in the shortest possible time by replacing fossil-based technology with renewable technology, cutting down greenhouse gas emissions, and stabilizing global temperature to <1.5 °C of pre-industrial levels. This is despite Franks, Keenan [1] revealing the conspicuous absence of critical minerals from the SDGs. At the current rate of advancement in technology, it is estimated that by 2050, 88 % of total global electricity would be generated from renewable energy sources from a 2020 baseline share of 29 % [2]. Critical

minerals have no definite definition, instead, they are any minerals that have essential function in contemporary technology and have “few or no substitutes” [3]. Hayes and McCullough [4] define it as any “valuable constituent element of a mineral commodity that is subject to the risk of supply disruption and which serves a purpose deemed as important based on the evaluators' perspective”.

In analysing governance risks in critical minerals, the question ‘to whom is a mineral critical?’ must extend beyond state and industry actors to include the perspectives of local communities. The paper argues that the stability of critical mineral supply chains increasingly depends on securing social license to operate at mining sites, positioning sub-national actors as central to the governance and geopolitical calculus of criticality. This argument is particularly salient in the context of the energy transition, where diversifying supply chains for critical minerals is imperative to reduce global vulnerabilities and ensure sustainable pathways for low-carbon technologies. My argument is that social license adds another layer of criticality as mining communities can

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withhold or acquiesce it to leverage negotiations in their interest.

My conceptualisation of social license to operate is similar to, but not the same as, the social and environmental risks that mining companies must address with their environment, social, and governance (ESG) programmes. Instead, it aligns with what Boone [5] refers to as '*entente*': the recognition by 'powerful social actors' (companies and investors) that their investment and operations can be sabotaged by ordinary people (community members) through collective resistance, malicious over-compliance (work-to-rule) or studiously enacted incompetence; unless they reach a mutual, often unspoken or unwritten, social contract – the social license to operate. '*Entente*' also implies that communities equally benefit from a business that they offer a social license for: any disruption or delays to the mining business in their community is equally detrimental to their interests.

The paper uses primary data collected in lithium-rich communities in Ghana to analyse the interrelationship between social license and global supply risks of critical minerals, considering the imperative of diversifying supply chains for the energy transition. Drawing on Ghana's nascent lithium sector, the study finds that social license is pivotal for mitigating supply chain disruptions, complementing technological pathways aimed at reducing mineral criticality [6,7]. The analysis highlights three core insights: first, mining agreements that reflect local priorities are essential for legitimacy; second, timely and inclusive civic participation strengthens trust and secures social license [8]; and third, mining communities are heterogeneous, requiring context-sensitive governance approaches. These findings underscore that representative participation and multi-actor coordination – spanning companies, government, community residents and traditional authorities – are central to embedding social sustainability in critical minerals governance and ensuring resilience in global supply chains.

In the next section, I critically reflect on the nexus between social license and the supply chains of critical minerals, followed by the methodology and an analysis of the case of Ghana, followed by conclusion and recommendations.

2. Revisiting criticality: whose needs, whose risks?

2.1. Supply chain risks and criticality in the Global North (industrial nations)

A fundamental question that has remained unsettled in critical minerals discourse is: whose perspective is defining the criticality of some minerals? The urgency of transitioning from fossil-based energy to renewables is widely recognized as essential for the green energy revolution [9,10]. Franks, Keenan [1] argue that while minerals security is essential for sustainable development, the SDGs largely overlook sustainable mining and its social dimensions. Incorporating this perspective underscores the need for socially responsible mineral sourcing and proactive community engagement in mining regions, ensuring that the energy transition does not reproduce extractive injustices that negatively impacts mining communities but instead, advances social sustainability alongside environmental goals. Depraeter and Goutte [3] have interrogated the criticality question in terms of supply risk and economic importance to argue that industrial technological needs and the state's broader economics and political geostrategy define the criticality of critical minerals. Mining companies also contribute to the criticality conundrum through their investment priorities in the supply chain of critical minerals in response to governmental policies, modern technologies, national security, and evolving global appetite for the energy transitions.

Politically, in the last decade, countries and regions such as the United States [11,12], European Union [13], Australia, the United Kingdom [14], Japan, and India [15] have developed and varied their own list of critical raw materials, based on supply risk and technological and economic needs. This has further reinforced the scholarly and policy discourse of placing industry and the political state at the centre of the

criticality question. Recent examples of minerals considered critical include cobalt, graphite, lithium, and the 17 chemical elements called rare earth minerals necessary for magnets and everyday technologies like flat-panel Television sets [3]. The UK, for instance, identifies both graphite and lithium as 'high criticality'. This means that these minerals have high global supply disruption risk and the UK sees them as integral to its technological futures: any supply chain disruptions would disrupt its economy and innovations industry [14]. For the six critical minerals that GLOBSEC [6] identifies as strategic to Western countries, the following striking statistics apply¹: China dominates over 70 % of global processing or refinery of these minerals, and in specific terms, processes or refines manganese, rare earth elements and natural graphite by up to 98 %, 87 % and 100 % share globally, respectively. Thus, any geopolitical crisis such as recent US-China Trade War [16,17], would have catastrophic consequences on western economies and technological innovations. Fig. 1 presents a visual map of the degree of criticality of some critical minerals based on a synthesis of current key country- and region-specific lists.

The dominance of a few countries like China in the value chains and supply chains can certainly sway global supply of these minerals for geopolitical and political purposes [6,12]. This creates a risk for the disruption of economies, technological innovation, and military geostrategy. Two-fold complementary pathways to minimising this risk to global supply chains include: (i) focusing "on innovation that minimizes critical minerals requirements" in modern technology [7] and (ii) diversifying supply sources of the most important minerals to reduce reliance on China; this necessarily implies a consideration of the interests and voices of Global South countries in criticality discourses [13,14].

For pathway (i), if research, innovation, and funding successfully scale up alternative battery technology being piloted or offer adequate incentives for the demand and supply of energy efficient technologies that rely on low critical raw materials [7], then the criticality focus on existing critical materials would likely shift in the next decade. Alternative battery chemistries such as Sodium-Ion (NA-ion) batteries are being widely used in China as alternative to lithium and performs better and safer than lithium in low temperatures at lower costs, although its lower energy density makes them less ideal for weight-sensitive and high performing applications like drones and electric vehicles [18]. Others include Lithium-Sulphur (Li-S) batteries that uses sulphur, which is in abundance with higher energy density but with shorter cycle life [18], and all-Solid-State batteries that have higher energy density, improved safety and faster charging but extremely expensive to produce [19]. Some are a few years from commercial production and would likely change the 'nomenclature' of critical minerals in the coming years. But this pathway requires strategic investment, foresight and state-led financing since most of the associated research and innovations do not always look intuitively viable, commercially.

2.2. Situating the social license to operate in the literature

Pathway (ii) requires ensuring that all minerals-rich countries and mining communities are partners of value in the critical minerals value chain to reduce risk of disruption and ensure sustainable supply. To reduce supply reliance on China and a few other countries, every country that has significant deposits of minerals such as Bolivia and Chile (lithium), Democratic Republic of Congo (Cobalt), Indonesia (Nickel) and South Africa (Manganese) should be considered partner of value. Criticality would therefore mean taking social license seriously.

Local community's awareness and utilisation of their social license can go either way (either undermine or support the mining industry's objectives) depending on the degree of community engagement and

¹ These are graphite, manganese, rare earth elements, copper, nickel and lithium.

2 nd Quadrant	Near Critical Minerals (Low supply risk + High centrality to energy transitions)	Critical Minerals (High supply risk + high centrality to energy transitions)	1 st Quadrant
	• Copper • Electrical Steel • Germanium • Silicon • Strontium • Uranium	• Antimony • Cobalt • Dysprosium • Gallium • Graphite • Iridium • Lithium • Magnesium • Neodymium • Nickel • Palladium • Platinum • Praseodymium • Rare earth elements • Silicon Carbide • Tellurium • Tin • Vanadium	
3 rd Quadrant	• Magnesium • Phosphorus • Rhenium • Titanium	• Aluminium/ Bauxite • Fluorine • Tellurium • Manganese	4 th Quadrant
	Important for the transitions but not normally considered critical (Low supply risk + Importance to transitions).	Elevated Criticality (High supply risk + Low importance to energy transitions)	

Fig. 1. Criticality matrix of some critical minerals adopted from Depraeter and Goutte [3] and Lusty, Saw [14].

ownership [20]. Paler [21] has argued that any discourse on extractives governance that disregards the subnational stakeholders (i.e., at the village, community, district levels) overlooks the constructive and destructive power of such stakeholders. This is true even if the processing and marketing happens elsewhere. My caution here is that a social license to operate ought to be community-focused; it should not be conceptualised as a bargain for business since it is not a legal or economic license. The crux of a true social license is underpinned by Free, Prior and Informed Consent (FPIC), an instrument for recognizing rights: true engagement, ownership and mutual consent [22,23].

Miller [24] interrogated the nexus between economics and a population's psychosocial and political behaviour in the mining sector and found that people's attitudes at the subnational level vary substantially along the spectrum of geographic proximity and the distribution of benefits. He argued that even in the midst of any resistance, there is always the possibility that "positive externalities" will attract support from some members of society [24]. The variation in the distribution of the support is directly proportional to the degree of political trust. Miller's finding is that where there is social license, political trust is higher in mining communities than in non-mining communities or communities further away from mining sites. This, Millar argues, is due to the relative benefits that accrue more to residents in the mining areas than those who are further away and feel disenfranchised despite often feeling the social and economic impact of the extractive activities.

Mining also has higher risk to people, for several reasons: First, it occurs on land, requires significant amount of time for prospecting and extraction, and affects people's livelihoods. It is therefore more likely to interfere with livelihood activities and social life than other extractives such as petroleum. Second, mining has historically generated significant community level tensions due mainly to legacies of colonial intricacies

[8,25,26]. Indeed, the controversies generated by mining have increased exponentially in the last two decades. While less than 15 new mining conflicts were recorded globally in 2000, this number increased to about 280 by 2016 [25]. Third, a substantial share of global mining activities occurs in low- and middle-income countries, many of which are characterised by political instability and weak governmental oversight. In such instances, governments are "unable or unwilling to take responsibility to effectively manage [the] negative impacts" [27,28]. In 2018 alone, 24 % of disputes recorded by the Convention and Additional Facility Rules of the International Centre for Settlement of Investment Disputes (ICSID) related to extractives, political instability [29].

While I employ the concept of social license to frame issues of community engagement and FPIC in this paper, I do so with caution, acknowledging that the concept of social license does not provide a straightforward, universal or silver bullet understanding of governance challenges [23,30–33]. For instance, Owen and Kemp [23] have argued that the criteria of FPIC are more important for mining communities than interventions that miss those four criteria. Again, the assumption that community stakeholders can stand up to the task of granting or withholding the rights of a mining operation to exist at the same or similar level as national regulators appears to be overstated [31]. Finally, social license is a two-way traffic – an 'entente' – in which both companies and communities build a relationship underpinned by mutual expectations: the way managers of mining operations perceive and understand a community's legitimate expectations is as important as the community's own claim to legitimacy [31,32]. This caution shapes this research in two ways. First, social license is unravelled with a analytical lens rather than a prescriptive solution, focusing on how communities perceive and negotiate legitimacy. Second, my analysis interrogates the limits of social license by highlighting tensions between

community expectations and formal regulatory mechanisms, illustrating that it cannot substitute for enforceable governance structures.

2.3. Criticality, conflict risks and social costs in the Global South

Failure to secure social license to operate could lead to costly conflicts, project delays, increased security spending, and reputational damage [25,34]. Beyond the realms of political morality is the risk of conflict, protests or disruptions in mining areas that can affect the supply chains of critical minerals [14,35–38]. These conflicts have costs to critical minerals mining and their supply chains.

For instance, countries like Bolivia, Chile and Zimbabwe have significant deposits of lithium, which is often found in rural and peri-urban areas. Yet, the voices and interests of local communities – groups of towns, villages and districts where these minerals are found – have often not been prioritized enough to elevate their living standards and well-being, especially when these communities are displaced or their livelihoods impacted by mining [39–42]. In some cases, unheeded grievances around civic participation and inequitable distribution of mining rents generate confusion, anxiety or protests that affect mining operations [43–46]. In Chile, several days of protests in 2019 disrupted access to lithium mining sites and was estimated to cost “more than \$1.4 billion of losses to businesses in a week” [44, para 13]. Elsewhere, Franks, Davis [47] reports of how conflict led a mining company with capital expenditure between \$3 billion and \$5 billion to lose an estimated \$20 million in net present value per week due to delayed production caused by conflict. In Bolivia, home to one of the world's largest untapped lithium, protests by indigenous communities of their displeasure over mining negotiations led to years of delays in the commencement of lithium mining [41,42].

The cost of conflicts or disruptive protests is immeasurable and influences the criticality of minerals. This cost includes fatalities and physical or psychological harm, economic costs (e.g., destruction of infrastructure, reduced investment), social costs (e.g., displacement, erosion of trust) [48,49], political costs, weakened institutions, and human rights abuses [50]. Such cost is equally applicable, if not more, to situations of conflict where (critical) minerals mining are involved, especially due to the so-called ‘resource curse’ dilemma in the Global South [51–53]. Historically, ecological distribution conflicts have occurred when local communities resist mining projects due to environmental and livelihoods concerns, risks to health, and cultural losses [8,25]. But, as Özkaynak and Rodriguez-Labajos [54] show, disruptive protests do not always ultimately have negative outcomes of disruption and fatalities, but have also historically shaped environmental justice and governmental or corporate reforms. Whatever the outcome of conflicts and protests, there are social, environmental [47] and corporate [55] risks. These outcomes, altogether, contribute to the risk of supply chain disruptions. The risk can be reduced through trust building, transparency, accountability, and equitable distribution of rents from minerals mining [8,48].

Although Ghana's lithium sector remains nascent and commercial mining has yet to commence, exploration activities and prolonged delays have already produced tangible social impacts in affected communities. Local media reports document grievances related to land use, compensation, and uncertainty, signalling early strains on community-company relations [56–59]. These dynamics suggest that social license will be contested from the outset of commercial operations. This unique feature makes Ghana an especially compelling case for examining social license within the broader context of supply chain risk governance.

3. Study site and methodology

This study investigates how sub-national and economic considerations shape the risk governance and supply chains of critical minerals, particularly the role of social license to operate (SLO). Focusing on Ghana's emerging lithium sector, it adopts a qualitative design, guided

by a need to capture nuanced social, cultural, and political dynamics in mining.

Ghana was chosen as the case study because of its emerging role in the global energy transition and the socio-environmental challenges linked to lithium exploration and the wider mining ecosystem [57,60–62]. Lithium was discovered in commercial quantities in the Mfantseman Municipality of the Central Region of Ghana in September 2023, and although communities have already been impacted by the exploration, discovery and moratorium on socio-economic activities, prolonged legal processes have delayed commencement of commercial mining [56–59]. Ewoyaa, Krampa Krom and Krofu are three of the affected communities. Ghana has an estimated 30.1 to 35.3 million tonnes of lithium deposit of 1.26 % lithium oxide and is tipped to be among the top 10 global producers of lithium [63,64]. Together with other minerals projects such as the Ghana Integrated Aluminium Development Corporation (GIADEC), the country's lithium will “become an important source of government revenue and create jobs and business opportunities” [64]. This emerging critical mineral sector draws comparison with other extractives [8,65]. The extractive industry in Ghana – gold, bauxite, diamond, petroleum – has had significant social, political, and environmental impacts [66–69]. Meaningful local, community engagement has suffered; lessons could be translated onto the critical minerals sector to reduce the risk of community upheavals and disruption.

Interviews with community members and key informants provided firsthand accounts of experiences of mining and risks to governance. Fieldwork conducted in August 2024 (see Fig. 2) included semi-structured interviews and focus group discussions involving local leaders, ordinary residents, and youth that have interest in the mining activities. Three focus group discussions (FGDs) were conducted involving: six female residents of Krofu, six male community leaders of Krofu, and seven male and female traditional leaders of Ewoyaa. The individual interviews comprised of (a) seven key informants with leadership in the local assembly and the surrounding communities (three females, four males) and (b) six in-depth interviews with ordinary residents of the three lithium-rich communities. This brought the total number of people interviewed to 32. Interviewees were asked to reflect on their experiences of lithium mining governance, expectations, participation, and degree of trust in the process. Interviews were transcribed, anonymized, and analysed using themes related to trust, satisfaction, legitimacy, participation, and resistance. Although the number of interviews conducted is small, the findings remain valid for the insights being claimed. Stake [70] emphasizes that the strength of qualitative research lies in its capacity for depth rather than breadth, allowing researchers to explore phenomena intensively without requiring large sample sizes. Similarly, the decision to focus on a single-country case – Ghana – rather than multiple cases is supported by established methodological literature. Yin [71] argues that a single case study, when it is unique or offers novel perspectives, constitutes a robust and powerful approach for generating new knowledge.

The literature and policy review offered contextual and regulatory insights. It situates the study within debates on critical minerals governance, social license, and community participation, examining how and why community engagement is necessary for critical minerals governance. Using databases such as Scopus and Web of Science, approximately 90 peer-reviewed articles and policy papers were selected through keyword searches (e.g., “critical minerals,” “social license,” “community mining governance”). The review identifies how different perspectives are represented – or excluded – in dominant criticality frameworks. Key policy documents, including the Minerals and Mining Policy, the Energy Transition Framework, and the 2023 lithium agreement, were analysed to assess how community engagement is institutionalized in national governance structures. To capture public narratives and elite discourse, local media content was analysed (e.g., Joy News, Citi Newsroom). These included interviews, news reports, and documentaries featuring government officials, corporate

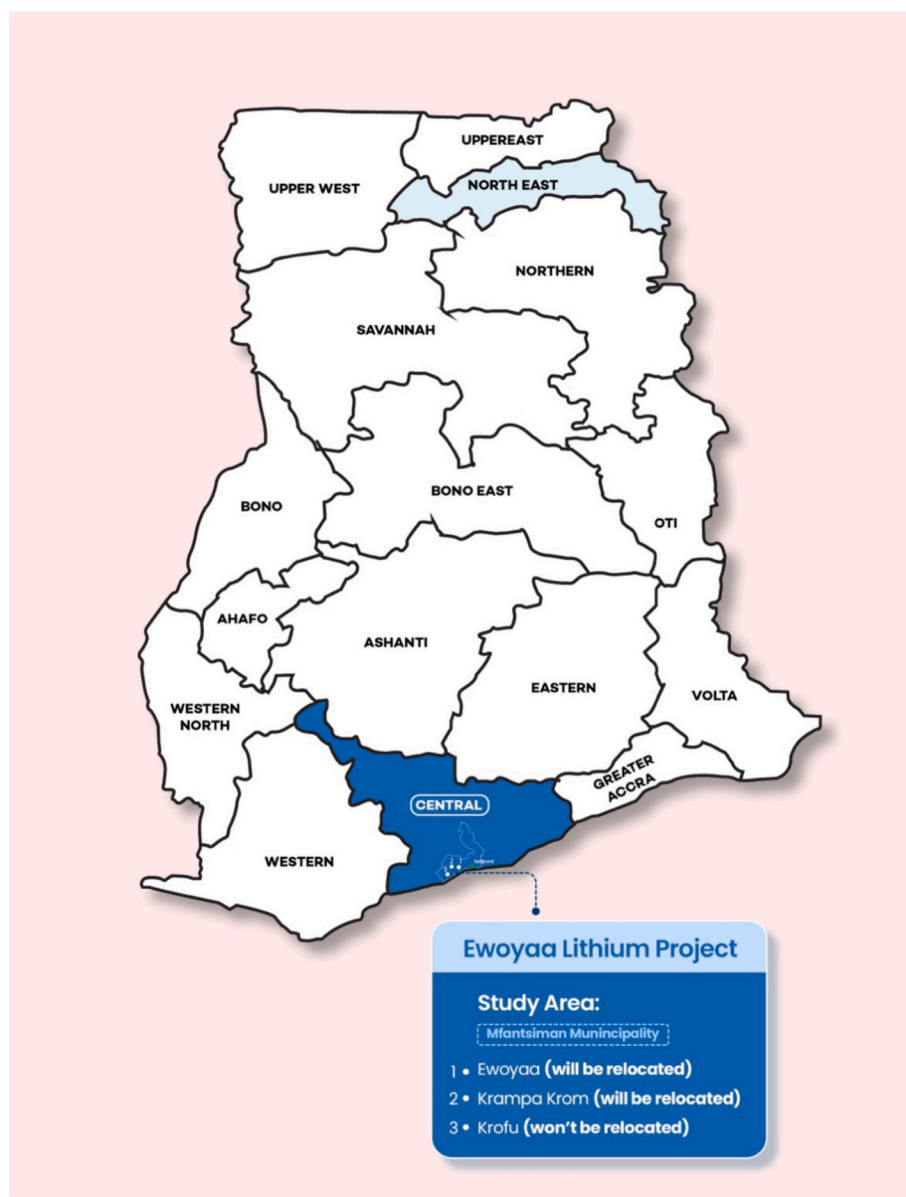


Fig. 2. The map of Ghana showing the ten regions of Ghana and the study site.

actors, and community leaders. Materials were transcribed and thematically coded to identify how social license and community concerns are framed in public discourse.

Thematic analysis was done by identifying recurring themes and trends of experiences, perceptions, and attitudes towards mining and stakeholders of mining. Direct quotes and tables were generated to distil and provide evidence for the themes. My positionality is reflexively considered throughout the research process. As an external researcher (non-resident), access to communities was facilitated through local gatekeepers and community leaders, ensuring trust and transparency. Sensitivities surrounding interviews with youth and women were managed by adopting culturally appropriate protocols, including obtaining informed consent and conducting interviews in safe, familiar settings. Gender-sensitive approaches were prioritized to avoid reinforcing power imbalances. Reliability and validity were addressed through careful sampling, triangulation of data sources, and verification of emerging themes through cross-referencing. This approach is particularly suited to unpacking how social license operates as a governance risk mechanism in critical mineral development.

4. Ghana case study

4.1. *The Lithium Mining Agreement differs from everything else – the best bet for social license?*

The Government of Ghana is not oblivious to the weaknesses in previous mining agreements that have historically undermined both governmental and community interests. It argues that such contractual anomalies have been addressed in the recent lithium Mining Agreement signed with Atlantic Lithium – an Australia-based private company that first discovered high-grade lithium in commercial quantities at Ewoyaa and its environs in Ghana's Central Region in 2023 [60,72]. Atlantic Lithium is advancing the Ewoyaa Lithium Project, set to become Ghana's first lithium-producing mine, under a landmark agreement that reflects new national policies on green mineral exploitation [73]. The Mining Agreement includes several clauses designed to enhance the legitimacy of the company's operations in Ghana. The Minister for Lands and Natural Resources, who led the negotiations, has emphasized that this Agreement stands apart from previous mining deals in Ghana and from other lithium agreements globally, declaring:

Ladies and gentlemen, respectfully, let me forcefully state that this lease differs from all previous mining leases in many respects.

[74, minutes 2:44]

Five main reasons are advanced to justify this declaration. First, the Agreement includes requirement of a mandatory payment of 10 % royalties from mined lithium (up from 5 % in all previous agreements), which “is one of the highest for exploitation of any mineral across the world” [74, minutes 2:57]. If the company defaults on its financial obligations, the Government would seize and sell the minerals mined, the company's assets, or abrogate the mining agreement and enter into a new one with another company [75]. Civil society has criticized the government for not bargaining for more in light of higher bargains elsewhere in Africa [76,77]. Botswana, for instance, negotiated for 50 % incremental share in its diamond with De Beers between 2023 and 2033.

Second, the country has “secured a 19% state participation in this mining company with a requirement to scale it up to a minimum of 30% Ghanaian participation through listing on the Ghana Stock Exchange” [74, minutes 2:55–3:35]. This will ensure that locals can purchase shares in the mining companies if they have the capital and are interested. Ghana's Sovereign Wealth Fund has made investments to be at the 66th position in terms of shareholding register, and the mining company sees the alignment of the company's and Ghana's strategic interest in the mining project as positive and welcome situation [63].

Third, “for the first time in the history of our country, a mining lease contains provisions for the establishment of a refinery; [for processing lithium] that is value addition and beneficiation” [74 minutes 4:05–4:28]. If the mining company decides that it is neither realistic nor cost-effective to establish such a chemical industry to add value before export, any third party that is able to do so in Ghana would have the first right of refusal for the supply of lithium [78, minutes 4:55–5:05]. Whether lithium would be exported in its raw state as is done with gold and other minerals mined in Ghana, or value would be added before export as is aspired by the government, would remain an area of tension, especially since the mining company “obviously want to maximize what this can bring for the project as well” [63, mm 6:06]. What happens would further have impact on the company's social license in Ghana, generally, since job creation is important for many ordinary people. Fourth, carried and participating interest (CAPI), “which is the shares the state holds without having to pay for it, is 10%” [78, minute 2:46]. This means that the government would have a minimum of 10 % shares in the mining establishment even before it commits any fund.

Fifth, the lease negotiation was paused for several months upon the commercial discovery of lithium (until July 2023) to allow the ministry to develop a general framework for all green minerals mining in Ghana for central government's approval [see also 73]. Such pause allowed the state to reappraise its existing minerals mining industry and to ensure that the shortfalls are addressed:

The approved policy includes provisions for an increase in the royalties' rate for green minerals, an increase in state and Ghanaian participation in all green mineral operations to a minimum of 30 %, just like GIADEC [Ghana Integrated Aluminium Development Corporation] and GIISDEC [Ghana Integrated Iron and Steel Development Corp], just like bauxite and iron ore, to a minimum of 30 %. An enhanced local content and local participation, including listing on the Ghana stock exchange and value addition and beneficiation for all green minerals in our country.

[74, minutes 1:56–2:31]

This has meant that despite the upheavals from some civil society groups against the lithium mining agreement due to what they say are their dissatisfaction over the rents accruing to the state [76,77], the government has remained convinced that it has made the best deal for the country.

Residents of the lithium mining communities have shown a range of views about the agreement between the Government and the company,

shaped by their experiences, expectations, and levels of trust.

“Being fully employed and satisfied is the most important thing. So, if this can be achieved, I will trust. Without this, I will not; I will feel disappointed.”

[Male resident, Ewoyaa, Interview 2024]

“I am very excited/happy. When they started the exploration, I offered a hospitality services, mobilised local communities, and provided some employment. When they started the exploration, they trained some of the locals. They trained some of them, some are already in school.”

[Landowner in Ewoyaa, Interview 2024]

Delays in parliamentary ratification and lack of visible action have however eroded confidence. Expectations that the mining will bring infrastructure, jobs, and investment keeps the social license alive. But among some residents, especially those of Krofu and Krampa Krom, who are earmarked for relocation, there is deep scepticism as they feel that the agreement benefits elites and external actors more than local peoples.

“Now, we do not know the actual truth. We do not even know whether if the blasting happens, it will affect our life and health or not. We do not know the entire truth; they are deceiving our leaders and us”.

[Female resident, Krofu, Interview in 2024]

“Even as Parliament is discussing the bill, not even a single youth has been trained.”

[Assembly person, Krofu]

“We do not know what the agreement says, we are told it is good for us.”

[Male resident, Ewoyaa]

Thus, social license is based on degree of trust, satisfaction and expectations, not just on signed agreements, which is consistent with findings from Sefa-Nyarko, Sinclair [8] that shows that satisfaction is fundamental to trust – a psychosocial phenomenon that is not static but evolving. Trust for the agreement in the three lithium communities – Ewoyaa, Krofu and Krampa Krom – is contingent upon whether promises (e.g., compensation, resettlement, and job creation) are fulfilled.

4.2. *Timely, civic participation offers satisfaction and secures social license to operate*

A novelty with the lithium mining agreement is the inclusion of a clause that mandates the mining project to establish a Community Development Fund worth 1 % of the annual revenue in addition to the 10 % royalty to government. Though a small proportion of rents, this sets the tone for acquiring a social license to operate.

The company shall set aside one (1) percent of annual revenue for a community development fund pursuant to a Community Development Agreement that shall be signed with the communities impacted by its operations [75].

The traditional authority in Ewoyaa is “satisfied with the nature of the engagements and conversation” with the mining company. The Secretary of the lithium mining company also declared satisfaction with their modus operandi in the communities and declared they have “support from our local government area, our community around us” [63]. The traditional authority has described engagement as “far advanced” [79]:

Firstly, let me commend the government for being able to negotiate for the 10 % royalties from the mining business. We are happy about it. It will benefit the community, it would benefit Ghana as a country, and it will benefit the Fanti people in general.

[79, mm 00:26–00:47]

However, during fieldwork, whilst residents expressed satisfaction with the nature of engagement with the mining company, a local NGO and the Environmental Protection Agency, they couldn't say the same about the central and local government authority due to what they consider as inadequate engagements.

Whilst there was unanimity in the township of Ewoyaa (one of the closest town to be relocated), opinions were divided in the township of Krofu, which is a town that would not be relocated but would have some of their farmlands being relinquished for mining. Krampa Krom would be relocated like Ewoyaa, but they appear more supportive of a waste management business in their locality than the mining business. Respondents expressed divergent satisfaction with different leadership stakeholders (e.g., traditional authority and local assembly persons) based on their expectations of them and the extent to which they believe those expectations have been met. Table 1 below shows the general community satisfaction comparison table.

Residents gave mixed messaging about their participation in decision making: some residents are actively involved in committees and meetings, while others feel excluded, especially those not directly affected by land acquisition despite their lives being impacted directly and indirectly through exploration and prospects for mining. A resident retorted: "The government has given out our land to them, so they will go ahead anyway" (Female resident, Krofu). The chiefs have previously expressed concern that the 1 % allocation for community development fund (CDF) is inadequate and would prefer something between 3 % and 5 % [79]. This is indicative that even the traditional authorities were not adequately consulted in establishing the threshold for the community development fund.

The biggest disappointment is regarding the moratorium or suspension of all community activities (e.g., farming, construction, economic activities) that tends to freeze the land area already in use by residents. Those whose lands have deposits of lithium can also not use their land: "Suddenly, they asked us to stop farming on the land for them to do their mining business" (Female resident, Krofu). This is consistent with the way the Legislative Instrument for Minerals and Mining (Compensation and Resettlement) Regulation, 2012 (LI 2175) describes an "inhabitant" and its obligations when minerals are discovered on their land:

"Inhabitant' means an owner or lawful occupier whose permanent dwelling or other immovable property is affected by a proposed mineral operation prior to the declaration of a mining area or a moratorium on further developments by the holder of a mining lease."

[80, Section 16]

The uncertainty caused by delays in parliamentary ratification of the mining agreement has made this moratorium very frustrating for residents. Even as there are mixed perspectives about the lithium mining itself, there is a sense that community voices are missing in the national level conversations. During interviews with youth leaders in Krofu (August 2024), they showed me a draft letter they intend to share with the media and other stakeholders to indicate that they are willing to make constructive inputs to the process, leaving their differences aside, if only Parliament could hasten the legal process. The 7th Parliament (2021–2024) could not ratify the mining agreement, however, which means that the disenfranchisement and discontentment remain a part of the lithium governance process. A traditional authority in Krampa Krom was unequivocal:

"We feel frustrated by the hold of our lives for the mining future, which we do not know. We know that there will be businesses, but we are not happy about this. We do not know what tomorrow brings, or what things will look like. We do not even know how things will be, but we have been asked to put our lives on hold"

[Traditional authority, Krampa Krom]

The social license to operate appears fragile as a result. Whilst some leaders and residents support the project, others feel coerced or inadequately informed, undermining legitimacy. There are notable similarities between the Ghanaian case and that of other lithium-rich countries such as Bolivia,² particularly around delays, lack of (or inadequate, late) consultation, and the suspension of livelihoods due to the lithium mining project [42,81,82]. For instance, in May 2025, the residents of Colcha K in the Nor Lipez Province halted lithium deals that the government was discussing with Chinese and Russian firms, citing unauthorised drilling and drying water sources [81]. A regional indigenous community leader explained that this is due to the lack of consultation despite indigenous livelihoods being significantly impacted:

"We've already seen wells drilled, springs drying up and our llamas migrating for lack of water..... We want the contracts suspended until they respect our right to be consulted."

[81, para 9]

Parliamentary debate over the lithium contract in Bolivia was eventually suspended due to such public outcry, and a court later ordered a halting of the agreement, citing lack of consultation and environmental safeguards [82].

In relative terms, the lithium mining company in Ghana has made more efforts in community engagement than the government, and in the very least, learns lessons as it goes. It has earned it legitimacy and some degree of social license. Residents would prefer the lithium mining company to manage their share of the 1 % CDF, which is a sign of social license. The communities have objected to any possibility of the CDF being disbursed centrally through the Government of Ghana's Consolidated Fund.

The consolidated fund that they want to put that 1 percent into, we, Nananom [traditional leaders of the area] disagree. We want that to be put into our fund to be managed by us and the company for easy access to the revenues. This will help us implement our development plans easily. If we get 5 percent, we will like it. [Kwansa II via 83, para 12]

Such early engagement by the mining company offers scope for social license to operate in the mining communities.

4.3. ...But mining communities are not homogenous

There are some sticky ends involving the mining company's engagement, however, including concerns by some youth leaders in Krofu – a community whose farmlands is earmarked for mining but not its residential areas. Residents argue that there is information asymmetry that disenfranchises them in decision making and participation [84]. Some farmers in the area have also claimed that their farms have been destroyed during prospecting of lithium without adequate compensation. A female farmer has for instance narrated in an Oxfam Alternative Mining Indaba: "The lithium company came onto my land without my permission..... They offered me basically nothing for what they destroyed" [85]. Another told me during interviews that compensations were made for undertaking exploration activities on her land, but these were inadequate to warrant the hassle she went through to receive compensation.

"Initially, I got GHC200 for two of us to share. Then later, they gave me GHC35 for the destruction of my farm produce. It upset me so much that I regretted even for chasing up on them. They destroyed farm produce of about a quarter of an acre".

[Female resident, FGD, Krofu]

Similarly, in Krofu, some local assembly members felt

² Bolivia has one of the largest lithium reserves globally.

Table 1
Community satisfaction matrix in the lithium mining communities.

Entity	Perceived responsibilities	Satisfaction level	Delineation of duties
Atlantic Lithium mining company	• Provide compensation for land and crops • Resettle affected communities • Create employment opportunities • Support local infrastructure (roads, hospitals, schools).	Mixed – Some praise for engagement and promises, others report disappointment and mistrust due to unmet expectations and lack of training or employment.	Expected to deliver direct community benefits and fulfil promises made during engagement
Government of Ghana	• Grant mining permits and licenses • Ensure fair compensation and resettlement • Oversee environmental and social safeguards • Communicate national policy and agreements	Low – Perceived as distant, slow in communication, and not engaging directly with communities	Seen as the authority granting mining rights and responsible for national oversight
Local Government	• Facilitate community engagement • Coordinate compensation and resettlement committees • Support local development planning	Low to moderate – Some involvement in committees, but many community members feel excluded or unsupported	Expected to represent community interests and coordinate with company and national government
Traditional Authorities	• Manage land and community representation • Negotiate with company on behalf of community • Ensure fair distribution of benefits	Mixed – Some trusted for transparency, others criticized for lack of communication or perceived collusion	Expected to safeguard community interests and mediate between stakeholders

disenfranchised, among others. But others express satisfaction with the compensation schemes and processes:

“They compensate us well. When they come on our land to dig or disturb the plants, the biological assets, they compensate us at premium value/ rate..... For me, we have not been treated badly. I can say that we have not received any bad deal, no company has treated us better than this company.”

[Male landowner, interview, Ewoyaa]

“I have received compensation after my cassava farm produce was affected by the drilling”

[Female resident, interview, Ewoyaa]

“So far, with the lithium company, they have done everything possible to deal well with us here, with all people.... Their approach is very cordial and respectful, so we have hope in them”

[Traditional leader, interview, Krofu]

Table 2 presents some of the nuanced attributes about satisfaction, trust and social license in the three communities, two of which are earmarked for relocation upon parliamentary approval of the mining agreement.

The stark difference in perspectives between residents in various communities have been explained variously. First, a Ghana-based

journalist, Samuel Bruce, who has reported widely about mining in the catchment area has explained that a potential cause of this ambivalence is that the mining company channels most of its community interventions and engagement to Ewoyaa, which has most of the lithium deposits, leaving behind other communities in the catchment area that have limited deposits such as Krofu, or that are further away from the main mining centres [86]. This phenomenon is consistent with findings from elsewhere [24], that the distribution of benefits from mining is more intense in mining communities than in adjacent communities, and that the farther one moves away from the actual mining sites, the less interested mining companies are to engage and offer socio-economic interventions.

A second explanation is that the term ‘community’ has ambivalent meaning and stakeholders are not homogenous [87]. The interest of community leaders does not always align with that of ordinary people. Neither do ambitions align among different community groups such as youths, adults, and occupational groups. This explains some of the information asymmetry between the traditional leadership and members of society found in the study for this paper. Information management is influenced by the priorities that people set. Further analysis of the interview data shows the risk of conflict or protest in the lithium mining catchment area; this has implications for social license and the success of the lithium mining project. It is directly proportional to the degree of

Table 2
Comparative analysis of community perspectives on lithium mining based on proximity to the mining site and possibility of relocation or otherwise.

Theme	Ewoyaa (TO relocate)	Krampa Krom ^a (to relocate)	Krofu (not relocated)	Comments
General Opinion	Generally optimistic: see mining as opportunity for jobs and development.	Mixed feelings: hopeful but frustrated by uncertainty and lack of progress.	Sceptical: feel misled and excluded from decision-making.	Ewoyaa is more hopeful due to engagement; Krofu more critical.
Satisfaction	Moderate satisfaction: some compensation and engagement reported.	Low satisfaction: delays and lack of clarity on relocation.	Low satisfaction: compensation seen as inadequate.	Satisfaction correlates with perceived fairness and inclusion.
Trust	Conditional trust: based on fulfilment of promises.	Low trust: feel promises are not being kept.	Very low trust: feel deceived by both company and government.	Trust is tied to transparency and delivery of promises.
Social License to Operate	Generally supportive if promises are fulfilled; expect benefits.	Supportive but contingent on relocation and compensation being fair.	Weak social license; feel excluded and misinformed.	Social license is stronger where engagement and benefits are visible.

^a Whilst Krampa Krom would relocate just as Ewoyaa, they have experienced the Zoomlion waste management company established on their land. According to residents, this company offers them better employment and socio-economic opportunities than the lithium mining would offer.

trust and satisfaction in the communities as presented earlier in Table 2 and is consistent with previous studies [8,25].

To assess conflict risk across the three communities, a qualitative conflict risk index is developed using a three-point scale: High, Moderate, and Low. This scale was chosen for its simplicity and clarity in capturing relative intensity of grievances and tensions based on interview data and media reports. Communities were classified according to indicators such as frequency of disputes, perceived trust deficits, and evidence of mobilization for protest. This approach provides a transparent and replicable measure of conflict risk for comparative analysis. Table 3 below offers a breakdown of these indicators for the risk of conflict, aggregated by proximity to the mining site and possibility or otherwise of relocation.

The potential for conflict or protests in Krofu and Krampa Krom is significant and could disrupt project timelines and community relations. Ewoyaa presents a more constructive engagement model. Early, inclusive, and transparent engagement is crucial for avoiding costly disruptions. Investing in trust-building measures such as training, compensation, and infrastructure is more cost-effective than managing conflict after it erupts.

5. Synthesis of findings

This section synthesizes the findings from the Ghana case study to provide analytical depth and connect them to broader debates on social license and critical minerals supply chain risk governance. While Ghana's lithium sector is still in its early stages, the evidence from this study reveals important dynamics that have implications for global supply chains. Mining agreements that align with local community needs, timely civic participation, and recognition of community heterogeneity emerged as central themes. These findings underscore that social license is not merely a conceptual ideal but a practical necessity for mitigating supply chain disruptions.

The community satisfaction matrix reveals that social license in Ghana's lithium sector is influenced by multi-actor responsibilities and uneven satisfaction levels. While Atlantic Lithium is praised for initial engagement, unmet promises regarding employment and training have generated mistrust. Similarly, the government and local authorities are perceived as distant and slow, weakening confidence in institutional oversight. Traditional authorities occupy an ambivalent position, trusted by some but criticized by others for lack of transparency. These dynamics underscore that social license is not negotiated solely between companies and communities; it is co-produced through complex governance relationships, as an 'entente'. This fragmentation amplifies

Table 3
Indicators of the risk for conflict in three lithium-rich communities in Ghana.

Community	Conflict risk	Examples from interviews
Krofu	High	Youth planned a demonstration (later cancelled under pressure). Assemblyman and youth fined by the Paramount Chief for voicing dissent. Accusations of exclusion from decision-making. Strong mistrust of the company and government. Threats to seize equipment if grievances are not addressed.
Krampa Krom	High	Deep frustration over halted livelihoods. Complaints of exclusion from negotiations. Concerns over land rights and lack of compensation. Strong mistrust of the relocation process. Sees mining as worse option compared with a waste management company on its land.
Ewoyaa	Moderate	Anxiety over delays and lack of information. Hopeful but cautious tone. Some trust in the process, but conditional on delivery of promises. Less confrontational than Krofu or Krampa Krom.

supply chain risk, as unresolved grievances at the local level can escalate into operational delays and reputational challenges globally.

Connecting these insights to the two complementary pathways outlined in Section 2, the Ghana case primarily informs Pathway (ii), which emphasizes inclusive partnerships with mineral-rich countries and communities. While Pathway (i) focuses on technological innovation and alternative battery chemistries, such as Sodium-Ion and Lithium-Sulphur batteries [18,19], these innovations do not eliminate the immediate reliance on lithium and other critical minerals. Therefore, ensuring social license in mining communities becomes essential for stabilizing supply chains during the transition to net zero.

The findings demonstrate that failure to secure social license – through transparent agreements, participatory governance, and sensitivity to local realities – could result in operational delays, community resistance, and reputational risks that reverberate across global supply chains. This analytical lens offers an understanding of how localized social dynamics intersect with global risk governance frameworks. By situating social license within supply chain analysis, the paper advances the argument that social sustainability is not peripheral but integral to criticality debates in the energy transition.

6. Conclusion and policy implications

This study analysed the interrelationship between social license and global supply risks of critical minerals, considering the imperative of diversifying supply chains for the energy transition. By situating Ghana's nascent lithium sector within this global context, the research demonstrates that social license is not a peripheral concern but a critical determinant of supply chain stability. The findings reveal that governance arrangements and community dynamics such as unmet expectations, participatory gaps, and heterogeneous interests, shape the prospects for operational continuity and risk mitigation. These insights underscore that the technological innovation alone, as outlined in pathway (i), cannot secure supply chains without complementary efforts under pathway (ii) to embed social license principles in resource governance.

The two policy pathways [6], which includes advancing alternative battery technologies and embedding social license principles in resource governance, remain central to mitigating global supply chain risks of critical minerals through diversification of sources. While technological breakthroughs such as Sodium-Ion and Lithium-Sulphur batteries offer long-term prospects for reducing reliance of certain minerals [18,19], they do not eliminate immediate dependencies on minerals such as lithium. Consequently, securing social license is indispensable for stabilizing supply chains for the transitions.

Timely community participation and effective governance structures emerged as pivotal in shaping community perceptions and trust. The findings show that social license is co-produced through multi-actor relationships involving companies, government agencies, and traditional authorities [25]. While mining companies' early engagement efforts were acknowledged [see other instances in 88], unmet promises and 'perceived' institutional distance have generated scepticism. Community dynamics further complicate this landscape, as mining communities are not homogeneous; they exhibit diverse expectations and socio-political realities. Recognizing these complexities is critical for designing governance frameworks that prioritize transparency, participatory decision-making, and equitable benefit-sharing.

If unaddressed, these issues could escalate into disruptive protests and social conflicts, revoking social license to operate and increasing the cost of conflict for the state, community members, investors [20,25,27]: financial costs (e.g., delays in operations due to protests), reputational costs (e.g., loss of investor confidence), and social and political costs (e.g., decline in trust, polarisation in mining communities). Effective risk governance goes beyond regulatory compliance. It embeds the principles of Free, Prior and Informed Consent, equitable beneficiation, and sustained community engagement [23,89]. The case of Ghana shows

that risk governance of critical minerals can be improved through sensitivity to local grievances about the mining, decision making, livelihoods and the environment. Balancing mineral development with social justice, responsible environmental governance, and long-term community empowerment is the imperative.

But there are shortfalls in the social license imperative. The social license to operate assumes an egalitarian purpose and a homogenous society that has common interest for sustainable social cohesion and environmental protection; this is false. Community involvement should include not just those that are closest to the site of mining, but also others who are several miles away and are affected by the mining through the impact on the economy, forests, water bodies, land, air, and general socio-economic and political contexts. Moreover, there are several risk areas that can undermine expectations of social license, including the possibility of co-optation, imbalances in power relations within communities and with companies, lack of synergy in community worldviews, and the silencing of local voices through global certification schemes due to their superimposition of standards [30].

The following recommendations for reducing risks and improving governance in emerging critical minerals sectors draw lessons from the current lithium project and community perspectives analysed in this paper. These recommendations aim to guide policymakers, industry actors, and future frameworks for critical mineral projects:

1. *National Policy Recommendations*: Policymakers should prioritize the development of a comprehensive critical minerals governance framework that embeds social license principles. This includes mandating early and continuous community engagement, transparent benefit-sharing agreements, and grievance redress mechanisms. Policies should also require independent monitoring of compliance and ensure that local development plans are aligned with mining operations to timeously respond to social discontent.
2. *Recommendations for the mining company*: To maintain its social license, mining companies should honour commitments related to employment, training, and community development. Companies should adopt participatory decision-making processes, provide regular updates on project timelines, and establish trust-building initiatives such as community liaison offices. Addressing unmet promises promptly and fostering transparency will be critical to sustaining community confidence.
3. *Future frameworks for critical mineral projects*: Future frameworks should institutionalize Free, Prior and Informed Consent (FPIC) as a non-negotiable requirement. They should also incorporate gender-sensitive and youth-inclusive engagement strategies, recognizing the heterogeneity of mining communities. Additionally, frameworks should integrate social performance indicators into licensing processes and link them to supply chain risk assessments, ensuring that social license is treated as a core component of global supply chain resilience.
4. *Environmental discourse communities*: This involves the formation of groups that share perspectives, beliefs and social goals around environmental and social risks to galvanise conversations between mining company and local community interests, emphasising the pros and cons of mining [90]. In this dialogue, mining companies could detail how they will manage the risk to the environment and society as well as how economic benefits will be distributed; local communities share their expectations, fears, and thorny areas of approval and disapproval.

By implementing these recommendations, Ghana and other minerals-rich countries in the Global South could strengthen governance in the critical minerals sector, reduce risks of operational delays, and position themselves as a reliable partner in global supply chains while safeguarding community interests.

CRediT authorship contribution statement

Clement Sefa-Nyarko: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Clement Sefa-Nyarko reports financial support was provided by King's Australia Menzies Institute. I consult for Participatory Development Associates in Ghana, but not directly related to mining. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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